RUTGERS

Semantics III

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Please pick up your **name card** from the front

office hours: Thursdays, 1:00-2:00 pm

Basement, Room 003, **Linguistics Building**

Homework 4, Part II, Question 3

does sentence (a) **entails** (b)?

(a) *Morgana **said she** studied for the test*.

(b) *Morgana studied for the test*.

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hence, if (a) is **TRUE**, (b) is **NOT NECESSARILY TRUE**, and therefore, (a) does **NOT** entail (b)

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hence, if (a) is **TRUE**, (b) is **NOT NECESSARILY TRUE**, and therefore, (a) does **NOT** entail (b)

Morgana SAID **she** studied for the test, but it does not mean this "**she**" is Morgana herself, even if this "she" actually studied

hence, if (a) is **TRUE**, (b) is **NOT NECESSARILY TRUE**, and therefore, (a) does **NOT** entail (b)

Previously on 201

tools to figure out meanings in Semantics:

for two propositions **p** and **q**:

entailment: a proposition **p** entails another proposition **q** iff, whenever **p** is **TRUE**, **q** must also be **TRUE**

presupposition: a proposition **p** presupposes aother proposition **q** iff both **p** and the **negation** of **p entail q**

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Set Theory

Set Theory: Basic Concepts

Set theory: Sets and Elements

a **set** is an abstract collection of items:

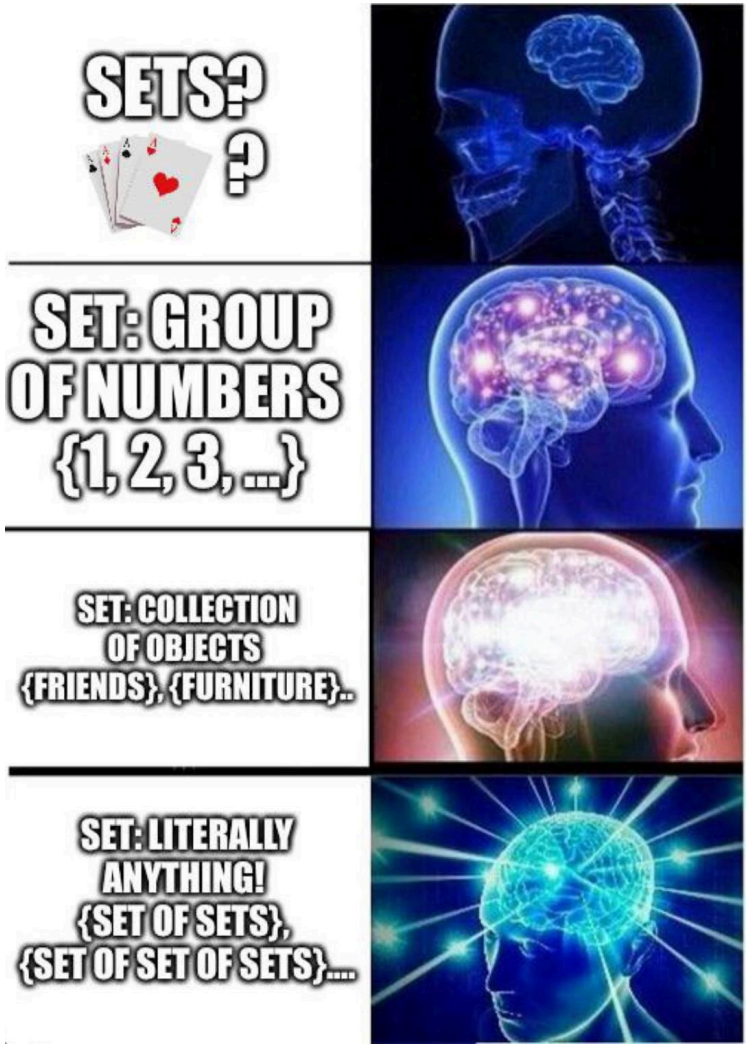
- {Mon, Tue, Wed, Thu, Fri, Sat, Sun} is the set of days of the week

the individual items in a set are called **member**s or **element**s:

- *Thursday* is a **member** or **element** of the set of days of the week

The symbols ∈and ∉ are used to specify whether an element is a member of a given set:

- *Mon* ∈ {Mon, Tue, Wed, Thu, Fri, Sat, Sun}
- *Oct* ∉ {Mon, Tue, Wed, Thu, Fri, Sat, Sun}



Set theory: Sets and Elements

- as in other branches of mathematics, we can use variables to stand in for values:

  A = {Mon, Tue, Wed, Thu, Fri, Sat, Sun} defines variable ***A*** as the set of days

- by convention, we use **capital letters (*A*, *B*, *C*...)** for sets and **lowercase letters (*x*, *y*, *z*...)** for elements

- we can first assign various elements to variables and then define sets based on them:

  a = Rutgers ***b*** = Stanford ***c*** = MIT ***c*** = Fudan

  - ***U*** = {Rutgers, Stanford, MIT, Fudan}
  - ***B*** = {Rutgers, Fudan}
  - [[**U**niversities]]
  - [[the **B**est Universities]]

- we can also define sets in terms of their **natural language meaning**

  - ***D*** = [[days of the week]]
  - [[Monday]] ∈ **D**

- we put [[ ]] around natural language expressions to show we're talking about their meaning, not their structure, morphology, or sounds

Set theory: Properties of Sets

- **order** of elements does **NOT** matter:
  - {a, e, i, o, u} = {u, o, i, e, a}

- **repeating** does **NOT** matter
  - an element is either **in** the set **or not**, it can't be in it twice:
  - {a, e, i, o, u} = {a, a, a, e, i, o, o, o, o, u, u}

- sets can be **finite** or **infinite**:
  - **finite**: set of days of the week = {Mon, Tue, Wed, Thu, Fri, Sat, Sun}
  - **infinite**: set of positive integers = {1, 2, 3…}

- members do **NOT** have to be **relevant** to each other
  - although in **Semantics**, they usually are:
    , 3, Wed, blue} is a legitimate set

in **mathematics**:

- a **SET** must be **well-defined**
- meaning we know how to decide **membership** clearly
- {2, 4, 6, 8, 10,...} (the set of even numbers)

yet in **semantics**:

- a **list** is also considered a **SET**

- {legitmate, 3x=5, OIIAI-cat.GIF, *never-gonna-give-you-up.MP4*}

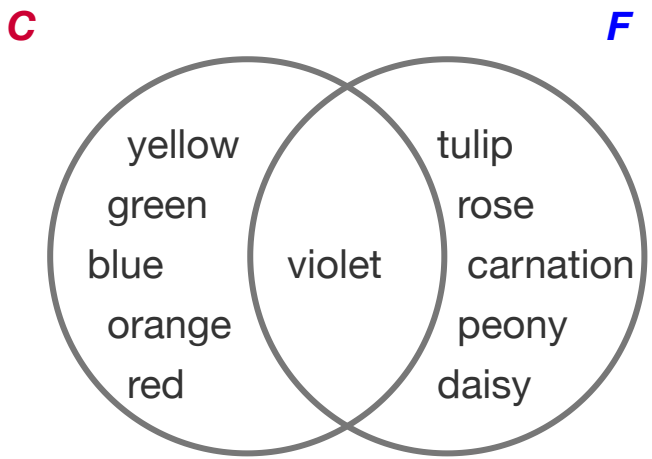




Set theory: Specifying a Set

- **list notation**: we can specify a set by listing its **members** between curly brackets
  - **C**(olours) = {red, orange, yellow, green, blue, violet}
  - **F**(lowers) = {tulip, rose, violet, carnation, peony, daisy}

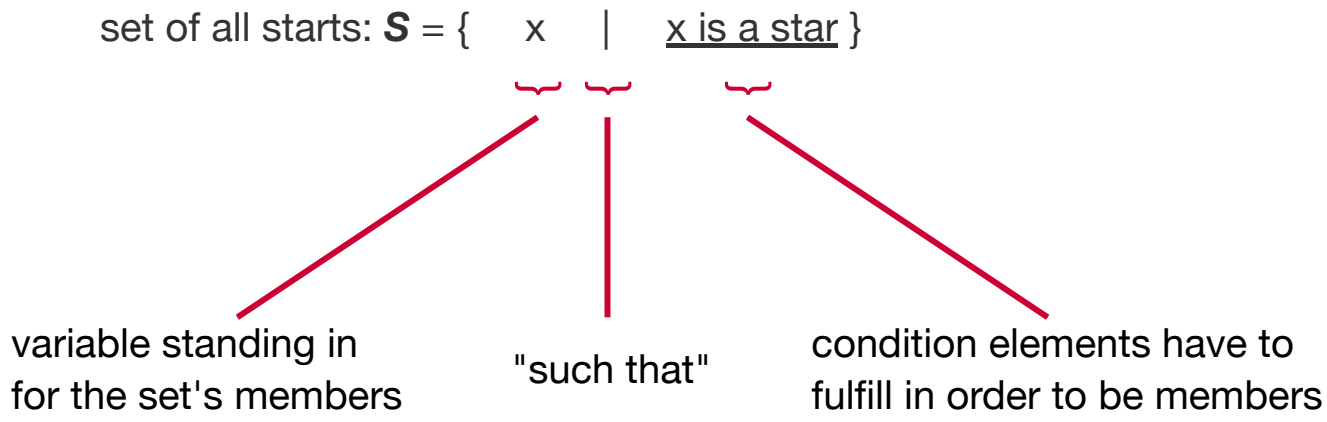
- **venn diagram notation**: we can specify a set by placing its **members** in a circle



Set theory: Specifying a Set

- sometimes we can't list all the members (we don't know them, there are too many)
- can anyone of us list **ALL** the stars in the universe?
  - Sun, Polaris, Proxima Centauri, Betelgeuse, Rigel...

- we can use **predicate notation** to define such a set:



- we read this out loud as: *S equals the set of elements x, such that x is a star*
- this is more or less the same as saying **the set of all stars** or the **set of all referent** of the word "star"

Words and Sets

- now that we know how to define a set, we can use it to define the meaning of a word
- recall that we're trying to define words in terms of their **reference**
  - the **set** of things they refer to
- we can define the reference of a word by:

list notation: [[bird]] = {goose, duck, sparrow, eagle…}

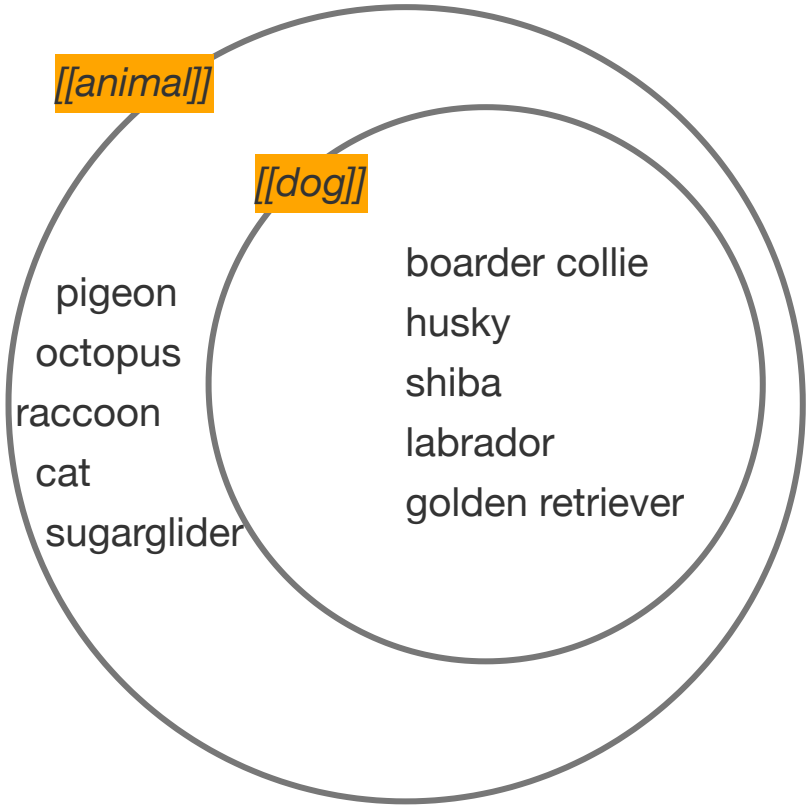
predicate notation: [[bird]] = { x | x is a bird }

- remember that we put natural language expressions like words in [[ ]]
  - this shows we're talking about the meaning of the word, not its structure

Set Relations

Sets

- the reference of words can **overlap** in various ways
- in order to **compare** two sets, we need **set relations
  - equality (identity)
  - subset
  - proper subset



Set Theory: Cardinality



- **cardinality** refers to the number of elements in a set (think cardinal numbers)

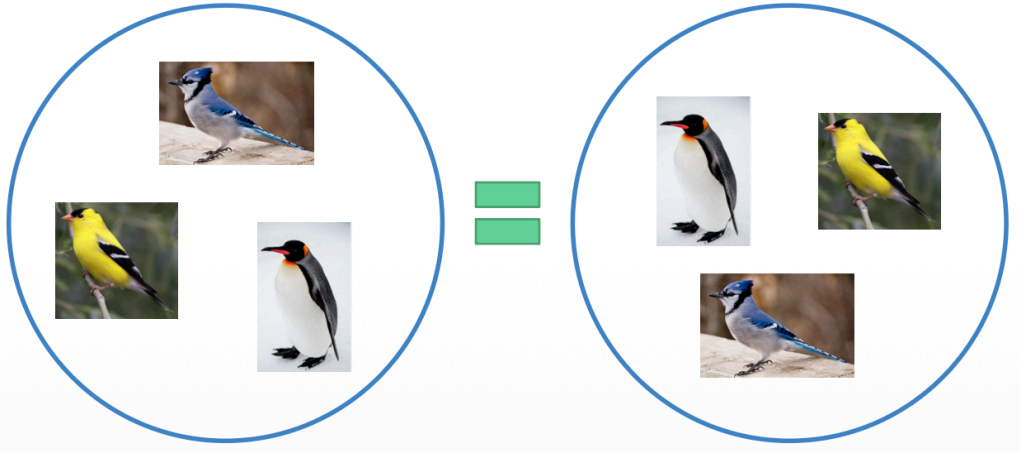
- we use vertical bars **|A|** to indicate cardinality
  - **|A|** is pronounced: cardinality of A, not absolute value of A)
  - suppose C = { red, orange, yellow, green, blue, violet }, what is |C|?

- a set with only one element (cardinality = 1) is called a **singleton set**:
  - {red}, {Wednesday}, {3}, {Snoopy}

- a set with no elements (cardinality = 0) is called the **null set** or the **empty set**:
  - an empty set is written as {} or Ø
  - but not as {Ø} (**why not?**)

Set Relations: Equality/Identity

- equality/identity symbol: $A = B$ (not equal/identical symbol: $A \neq B$)
- two sets are equal or identical if they have **exactly the same elements**:
  - { a, e, i, o, u } = { i, e, a, u, o }
  - { x | x is a bird } = { x | x is a winged, feathered animal that lays eggs }
- **formally**: $A = B$ **iff** for every x: $x \in A$ iff $x \in B$

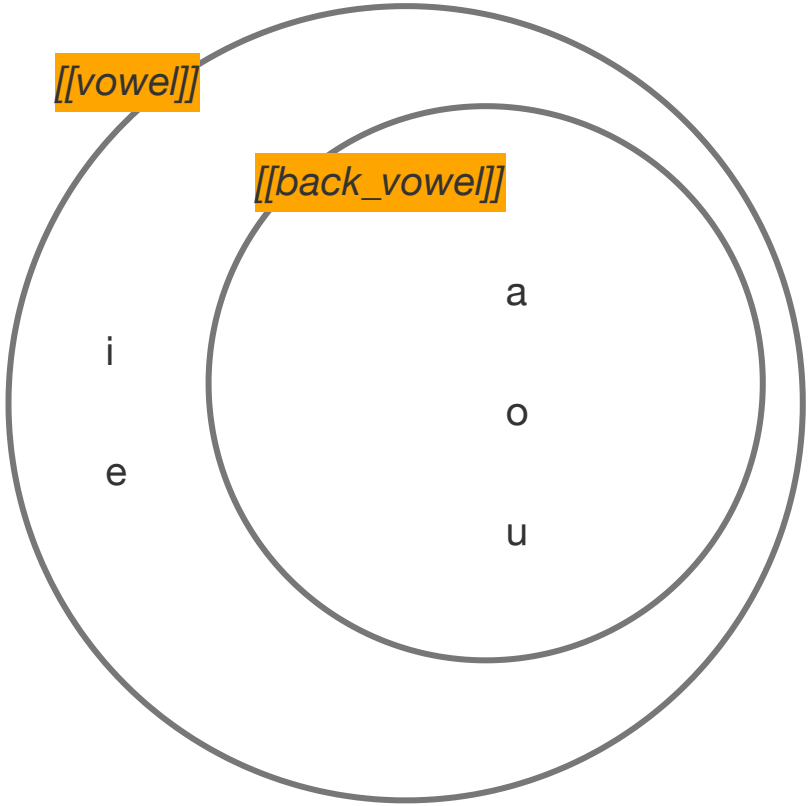


Set Relations: Subset

- subset symbol: $A \subseteq B$ (not a subset symbol: $A \not\subseteq B$)
  - A is a subset of B if **every element of** A is **also** an element of B:
  - { a, o, u } ⊆ { a, e, i, o, u }

- **formally**: $A \subseteq B$ **iff** for every x: if $x \in A$ then $x \in B$
  - if A contains **anything** that's **NOT** in B, it's **NOT** a subset of B:
  - { a, o, y } ⊈ { a, e, i, o, u }

- **note**: if A and B are identical ($A = B$)
  - A is technically still a **subset** of B, because every element in A is also in B:
  - { a, e, i, o, u } ⊆ { a, e, i, o, u }



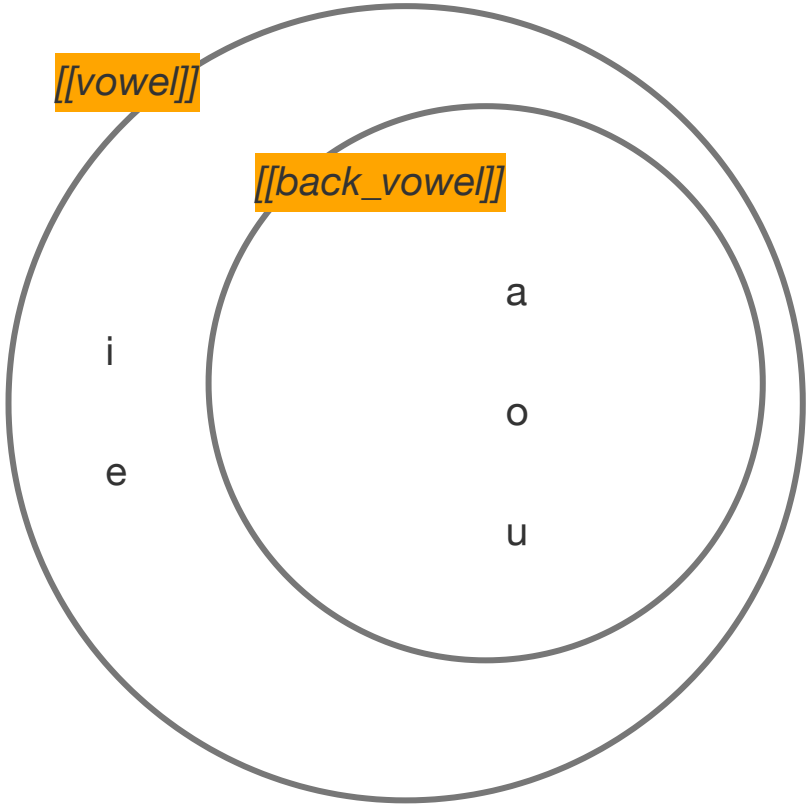
Set Relations: Proper Subset

- proper subset symbol: $A \subset B$ (not a proper subset symbol: $A \not\subset B$)
  - A is a **proper subset** of B if A is a **subset** of B **and** they are **NOT** identical ($A \neq B$)

  - $\{ a, o, u \} \subset \{ a, e, i, o, u\}$
  - $\{ a, e, i, o, u \} \not\subset \{ a, e, i, o, u\}$

- **formally**: $A \subset B$ **iff** $A \subseteq B$ **and** $A \neq B$

- in natural language, we **typically** use subsets ($\subseteq$) rather than proper subsets ($\subset$) because they're more flexible



Exercise I: Set Relation

are the following statements **true** or **false**?

1. B ⊂ A
   - ○
2. B = C
   - ○
3. B ⊆ C
   - ○
4. B ⊂ C
   - ○
5. B ⊆ D
   - ○

A = {Mary, Peter, Ivan, Josefina, Antonio}

B = {Peter, Ivan, Antonio}

C = {Antonio, Peter, Ivan}

D = {Peter, Mary, Ivan, Josefina}

Exercise I: Set Relation

are the following statements **true** or **false**?

1. B ⊂ A
   - **true**, all elements in B are also in A
2. B = C
   - **true**, order does not matter
3. B ⊆ C
   - **true**, if B = C, B is also a subset of C
4. B ⊂ C
   - **false**, to be a proper subset, there must be elements in C but not in B
5. B ⊆ D
   - **false**, there are elements in B that are not in D

A = {Mary, Peter, Ivan, Josefina, Antonio}

B = {Peter, Ivan, Antonio}

C = {Antonio, Peter, Ivan}

D = {Peter, Mary, Ivan, Josefina}

Set Relations in Natural Languages

Set Relations in Natural Language

- now that we know how to talk about **set relations** (equality, subset), we can return to our discussion of **synonyms**, **hyponyms**, and **hypernyms**

  - are the following pairs of words **synonyms** or **hypernyms/hyponyms**?
  - what is the set relation between them?
  - how would you write that out using **set theory** notation (=, ⊆/⊇)

cats vs. felines	[[cats]] = [[felines]]
drink vs. beverage	
squirrel vs. mammal	
sparrow vs. bird	
soft drink vs. soda	
beverage vs. juice	

Set Relations in Natural Language

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sparrow vs. bird	[[sparrow]] ⊆ [[bird]]
soft drink vs. soda	[[soft drink]] ⊆ [[soda]]
beverage vs. juice	[[beverage]] ⊇ [[juice]]

Principle of Semantic Compositionality

- recall that we can talk about two levels of semantic meaning:
  - **lexical semantics**: the meaning of words
  - **compositional semantics**: the meaning of sentences

- we can do the same with set theory

- **Principle of Semantic Compositionality** states that the meaning of **sentences** derives from the meaning of their **words** plus the meaning of their **structures**
  - we have known how to represent **words** using set theory
  - and we need to find a way to use set theory to show a sentence's **structures**

- we will only look at **declarative** sentences, and ignore **interrogative** and **imprative** sentences for now
  - since a declarative sentence is what we considered a **PROPOSITION** (or a statement)
  - a proposistion can be either **true** or **false**

(a) *Snoopy is a dog*.

(b) *Birds are animals*.

Declarative Sentences and Claims



declarative sentences typically state propositions in **two** steps

- first, they introduce some **topic** in the subject of the sentence

(a) *Snoopy is a dog*. (b) *Birds are animals*.

- then, they make some **comment** about that **topic** in the predicate:

(a) *Snoopy is a dog*. (b) *Birds are animals*.

so to translate into a sentence into set theory, we need to figure out how to **identify a sentence's topic and comment and then state it in set theory**

Compositional Semantics: Name is NOUN

Snoopy is a dog.

- what is the **subject** (**topic**)of this sentence?
- what is the **predicate** (**comment**) of this sentence?
- what is the **predicate** saying about the **topic**?
- is there anything in this sentence that could be represented as **a set** or **an element**?

Compositional Semantics: Name is NOUN

Snoopy is a dog.

in a sentence with the structure **Name is NOUN** like *Snoopy is a dog*

- the **subject** is a **proper noun** – that is, an **element**, not a set
- the **predicate**:
  - introduces some noun referring to a **set**: [[dog]] = set of all dogs
  - makes a claim about the **relationship** between the subject and this set: Snoopy is a **member** of the **set** of all dogs

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[[Snoopy]] ∈ [[dog]]

- in general, sentences of the structure Name is NOUN can be written as **[[name]] ∈ [[NOUN]]**

Compositional Semantics: Name is ADJ

***Snoopy is happy*.**

- this sentence has the form **NAME is ADJ**
- we can use set theory to define adjectives, too:
  - *happy* refers to the **set** of all happy things in the world, so **[[happy]]** = { x | x is happy }
- the sentence *Snoopy is happy*:
  - mentions an individual (**element**) Snoopy and a set [[happy]]
  - makes the claim (**comment**) that Snoopy is a member of [[happy]]

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[[Snoopy]] ∈ [[happy]]

- Sentences of the form Name is ADJ can be encoded in set theory as: **[[name]] ∈ [[ADJ]]**

Compositional Semantics: Name VERBs

***Snoopy runs*.**

- the sentence has the form Name VERBs – the **predicate** is a verb

- suppose we define the verb run as referring to the **set** of all things in the world that run:
  [[run]] = {x | x runs}
- then the sentence above
  - introduces an **element** Snoopy, introduces a **set** [[run]]
  - again makes the **claim** that the element is a member of the set

Compositional Semantics: Name VERBs

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[[Snoopy]] ∈ [[run]]

- sentences of the type NAME VERBs can be encoded: **[[name]] ∈ [[VERB]]**
- all three sentence types (**NAME** is NOUN/is ADJ/ VERBs) above can be encoded in the same way

Compo Semantics: Commoun Noun Subjects

***Cats are mammals*.**

- now the subject doesn't refer to an **individual**, it refers to a **set** of things: **[[cat]]**
- yet the **predicate** is the same as above sentences, which
  - introduces another **set**: [[mammal]]
  - makes a **claim** about the **relationship** between the **subject** [[cat]] and [[mammal]]
- can we say **[[cat]] ∈ [[mammal]]**?
  - **all** members of [[cat]] are **also** members of [[mammal]]

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  - **all** members of [[cat]] are **also** members of [[mammal]]

[[cat]] ⊆ [[mammal]]

- sentences of the type Nouns are Nouns can be encoded: **[[NOUN]] ⊆ [[NOUN]]**

Compo Semantics: Commoun Noun Subjects

Cats are fuzzy.

- *cats are fuzzy.* has the form **NOUNs are ADJ**
  - [[fuzzy]] = **set** of all fuzzy things = {x | x is fuzzy}

Cats meow.

- *cats meow.* has the form **NOUNs VERB**
  - [[meow]] = **set** of all things that meow = {x | x meows}

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[[cat]] ⊆ [[meow]]

note that we use **subset (⊆)** rather than **proper subset (⊂)**, because it works even when the sets in the subject and predicate are **synonyms**

[[beverage]] ⊆ [[drink]]

??? [[beverage]] ⊂ [[drink]]

- so using subsets gives us more **flexibility** and helps us deal with ambiguity in language

Compositional Semantics: Truth Values

- as we've seen, declarative sentences state some proposition, which can be encoded in set theory
- but just because you **state a proposition**, that does **NOT** mean it's **true**
  - you can make a **false** claim
- You can lie, you can be wrong, or you can say things without knowing if they're true
- so when we think of what a sentence means, we need to **separate** these two concepts:
  - what is **its proposition**? (and how do we **encode** that proposition in set theory?)
  - is the proposition **true** or **false**? (that is, what is its **truth value**?)
- defining sentences in set theory and determining whether they're true are two **separate** processes

(a) *Snoopy is a cat*	[[Snoopy]] ∈ [[cat]]	**FALSE**
(b) *Cats bark.*	[[cat]] ⊆ [[bark]]	**FALSE**
(c) *Dogs are mammals.*	[[cat]] ⊆ [[mammal]]	**TRUE**

Compositional Semantics: NPs

- suppose I wanted to use an **NP** like *yellow birds*

What kind of birds do you like?

Yellow birds!

- by using this **NP**, we are not making a **claim**
- we did not say "*birds are yellow*"
- we are just trying to refer to **all elements** that are **simultaneously** in both [[bird]] and [[yellow]]
- to define this in set theory, we need **set operations**

Set Operations

- **set operations** are ways of taking **existing sets** and creating a **new set** out of them
  this is analogous to how **addition** and **subtraction** take **two** numbers and make a **third** number

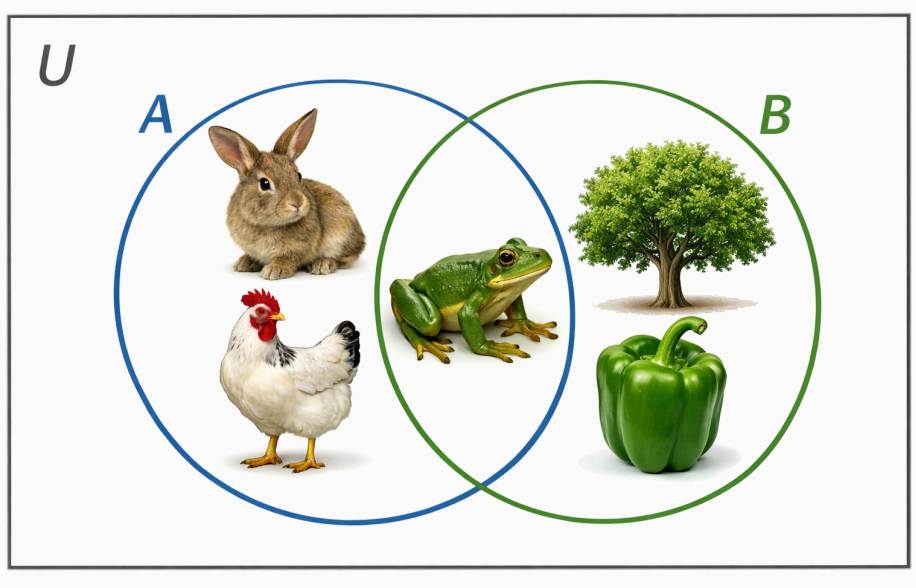
- four set operations: **union (∪)**, **intersection (∩)**, **difference (–)**, **complement (′)**

imagine a **universe U**, in that universe, there are five elements divided into two **sets**

- let A = { x ∈ U | x is an animal } = { rabbit, chicken, frog }

- let B = { x ∈ U | x is green } = { tree, frog, pepper }

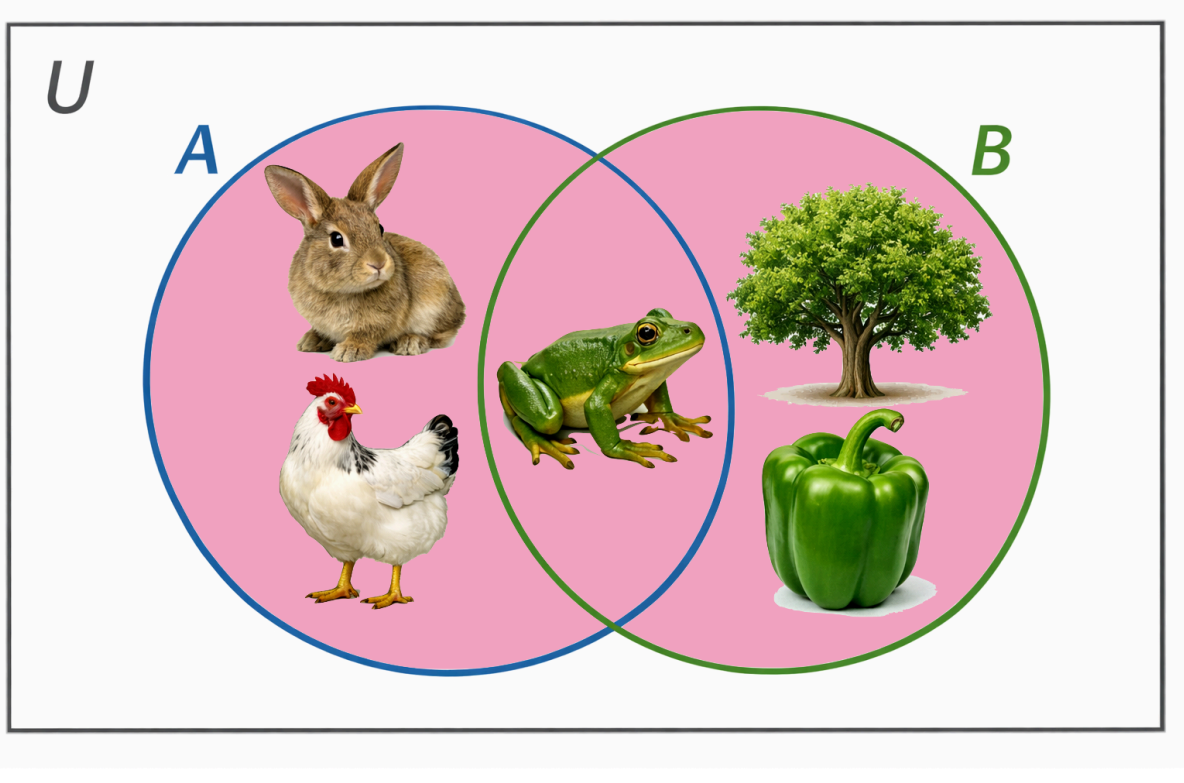
- notice that **frog** is in both A and B



Set Operations: Union

union of A and B produces a new set, which contains all **elements** that **belong to A or B** (or both).

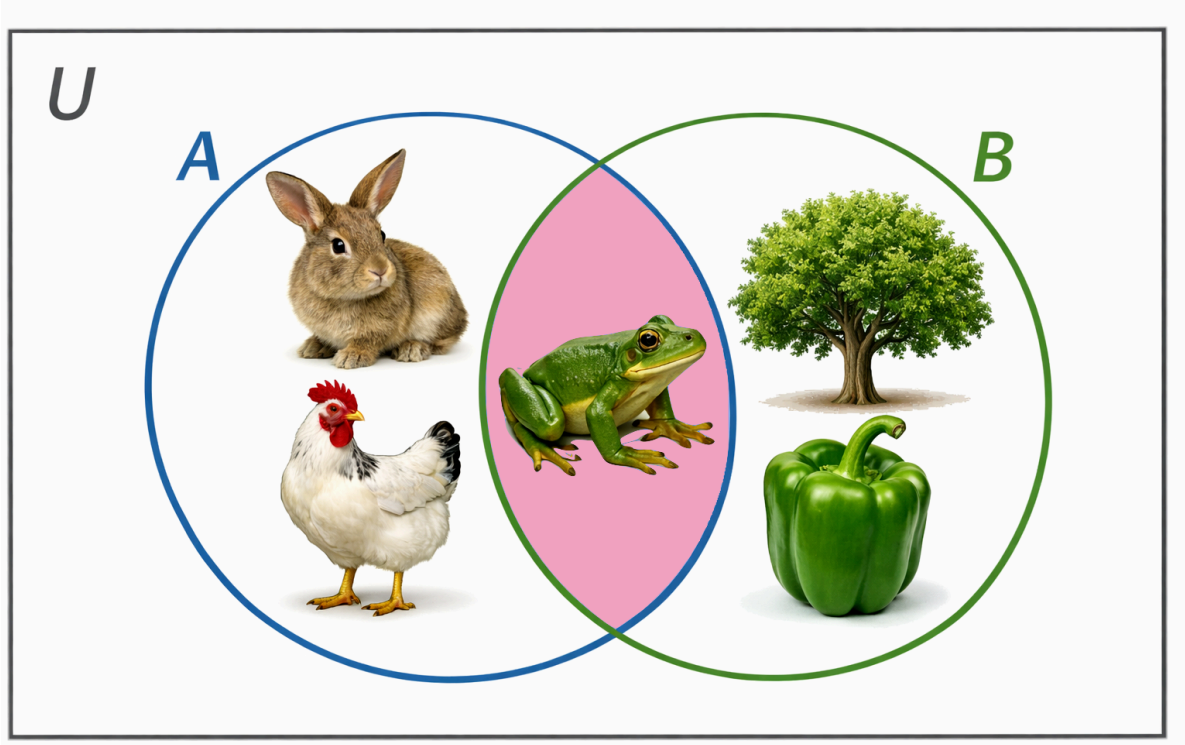
- A ∪ B = { rabbit, chicken, frog, tree, pepper }



Set Operations: Intersection

intersection of A and B produces a new set that contains all elements that **belong to both A and B**

- A ∩ B = { frog }
- **intersection** is the part that **overlaps** between two sets in a Venn diagram

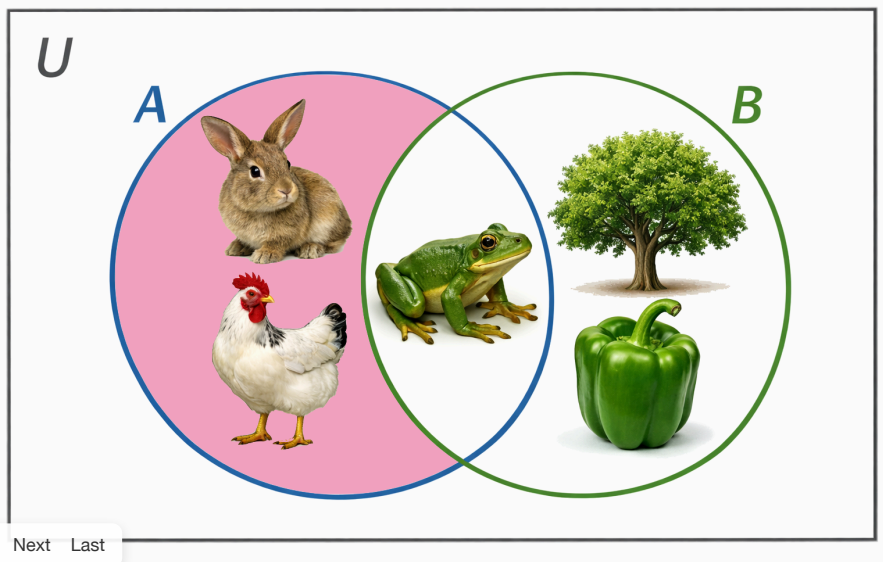


Set Operations: Difference

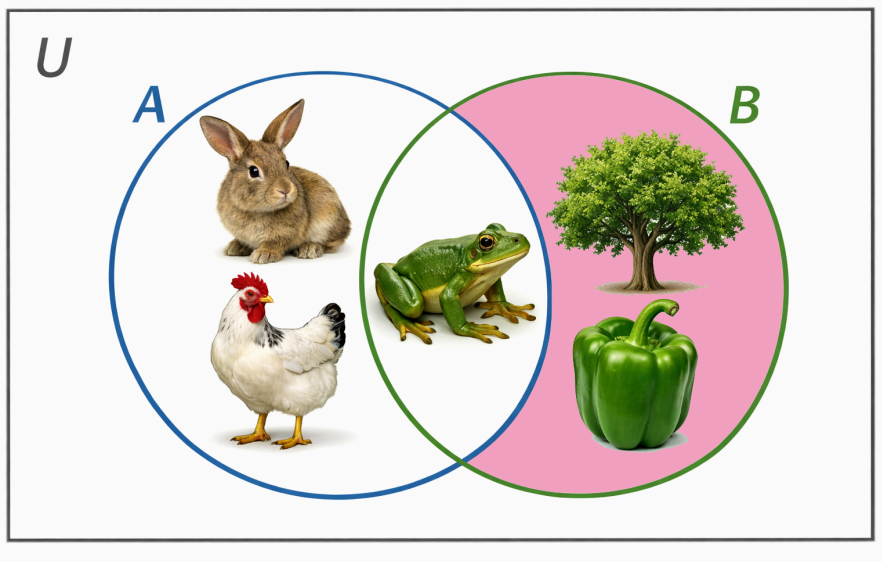
difference (–) of A and B produces a new set, which contains all elements that **belong to A but not B**

- A **–** B = { rabbit, chicken }
- **note** that A **–** B **≠** B **–** A

A – B



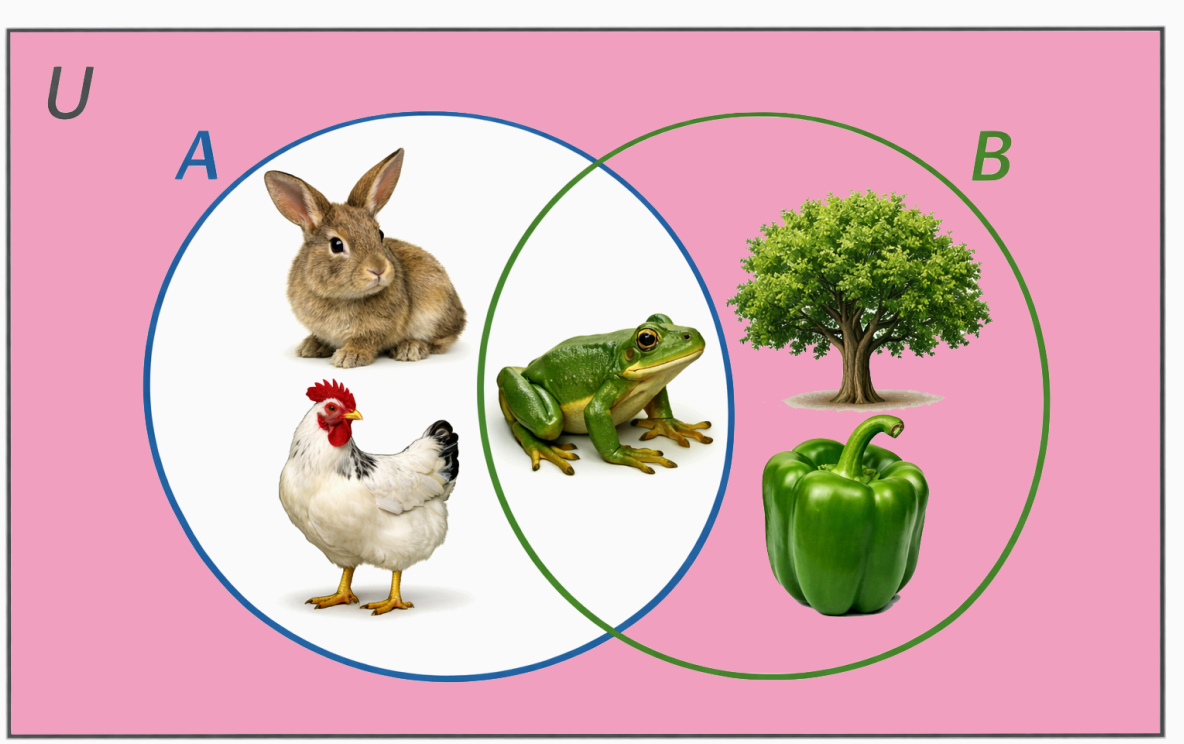
B – A



Set Operations: Complement

complement of A produces a new set, which contains all elements **in the specified universe (U) that are not in A**

- A' = { tree, pepper }
- the symbol of **complement** is tiny, so be careful

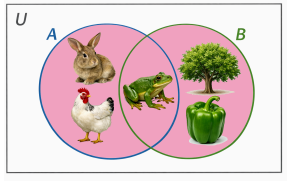


Set Operations: Summary

union (∪)

$A \cup B = \{ x \in U \mid x \in A \text{ or } x \in B \}$

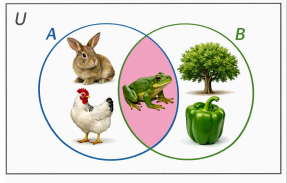
(elements in A or B or *both*)



intersection (∩)

$A \cap B = \{ x \in U \mid x \in A \text{ and } x \in B \}$

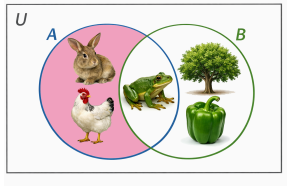
(elements in both A and B)



difference (–)

$A - B = \{ x \in U \mid x \in A \text{ and } x \notin B \}$

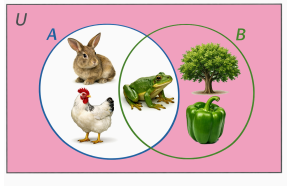
(elements in A but not B)



complement (')

$A' = \{ x \in U \mid x \notin A \}$

(elements in U but not A)



Exercise II: Set Operations

calculate the following formulas, and **name** the set operation(s) involved

1. A ∪ B

2. A – B

3. B ∩ C

4. B ∩ D

5. C′

A = {apple, pear, plum}

B = {orange, pear}

C = {plum, apricot, orange, nectarine}

D = {nectarine, apple}

universe of fruit: apple, pear, plum, orange, apricot, nectarine

6. (A ∩ B)′

Exercise II: Set Operations

calculate the following formulas, and **name** the set operation(s) involved

1. A ∪ B
   - **union**: {apple, pear, plum, orange}
2. A – B
   - **difference**: {apple, plum}
3. B ∩ C
   - **intersection**: {orange}
4. B ∩ D
   - **intersection**: Ø or {}
5. C′
   - **complement**: {apple, pear}

A = {apple, pear, plum}

B = {orange, pear}

C = {plum, apricot, orange, nectarine}

D = {nectarine, apple}

universe of fruit: apple, pear, plum, orange, apricot, nectarine

6. (A ∩ B)′
   - **intersection**: A ∩ B = {pear}
   - **complement**: (A ∩ B)'= ({pear})' = {apple, plum, orange, apricot, nectarine}

Homework V is published and is due **Mar 29** at **11:59 pm**

continue reading: the chapter **Semantics I: Compositionality**
from Victoria Fromkin et al.'s *An Introduction to Language*



Slides created via the R package **xaringan**.